

# Quest for alpha

Alexandre Furtado Neto

Independent Researcher

alexandre.com@yahoo.com

September 2, 2026

## Abstract

We continue the program of the toy universe: a deterministic, integer-only lattice automaton on a 3-torus from which geometry, a light clock, charge, polarization, and a sieved force emerge from a single bit-local rule. This paper executes a pre-registered search for the fine-structure constant  $\alpha$  from first principles: six candidate definitions, fixed in advance, are measured with a bit-identical harness and judged against  $1/137.035999177$  under a strict convergence criterion. All six fail at the accessible scales, and four structural properties of the dynamics are established along the way. The contribution is a complete, falsifiable negative battery—a methodological template for computational derivations of dimensionless physical constants.

**PACS numbers:** 12.10.Kt, 03.65.Ta, 04.60.Pp, 05.45.-a

**Keywords:** it from bit, digital physics, cellular automata, emergent spacetime, emergent charge, wavefront automaton, computational universe.

## 1 Introduction

In [1] we presented the *Toy Universe*: a deterministic cellular automaton on a 3-torus in which the light clock, charge, polarization, and the matter–antimatter asymmetry emerge from a single bit-local rule. This paper asks the sharpest question the program can pose: does the emergent coupling of that automaton reproduce the fine-structure constant from first principles? We answer with a controlled experiment. Six candidate definitions are pre-registered in Section 4.4—fixed before any comparison with the empirical  $1/137.035999177$ —measured with a bit-identical headless harness, and judged under a strict eight-digit convergence criterion.

All six fail at the accessible scales, and four structural obstructions are established along the way (the  $L = 7$  degeneracy of the polarization pair, the sticky two-bubble contact, the inexceedable maximum information speed, and the collapse-dominated character of the only force the model possesses). The failure set, not a number, is the result: in a domain—computational derivations of dimensionless physical constants—where numerology has been the historical norm, this paper demonstrates a complete, falsifiable negative battery as the disciplined alternative. Section 2 recaps the model; Sections 4.4 and 6 pre-register and measure the candidates; Section 7 draws the structural and methodological conclusions.

## 2 Recap: the it-from-bit automaton

In [1] the automaton is defined as follows. Space is a three-torus of side  $L$ ; a fourth coordinate  $w$  indexes  $W$  winding layers grouped into islands. Each cell carries six charge bits ( $c_0 c_1 c_2 q w_0 w_1$ ), integer radius fields  $r^2, r$ , and an affinity label  $a$ ; every update is integer-only. A pulsating spherical wavefront—the light clock—expands and contracts with period  $2R_{\max}$ , advancing one

cell per tick, and schedules the local dynamics. At each turnaround an election chooses a momentum direction, a helical walker broadcasts an arrival stamp, and every cell reconstructs a transverse polarization pair  $(pol_u, pol_v)$  approximating the circle  $u^2 + v^2 = R^4$  by integer square roots. The *s2B* sieve gate  $((u \cdot t) \bmod S) < u$  admits a fraction of overlapping active cells to the interaction branch; gate-passing contacts collapse and reissue. All quantities in this paper are measured under this single dynamics [1].

## 2.1 One scenario

The model defines exactly one scenario: the full deterministic dynamics of [1]—the transition function that the reference GUI labelled “full simulation.” The scenario index carried by the reference implementation is legacy debug scaffolding: in the current code it selects display colours and help text only, and never branches the automaton’s transition function. Every measurement in this paper (validation, candidates  $\alpha_A$  and  $\alpha_D$ ) is therefore taken under this single scenario. The two-bubble and canonical Platonic-seed arrangements of Section 6 are not scenarios; they are initial conditions, i.e. diagnostic probes, within the one scenario.

## 3 The electric channel as a measurable coupling

### 3.1 Force law of the automaton

The first contribution turns the qualitative Coulomb-like dynamics of [1] into a measurable law. In the interaction branch an electric contact occurs when either bubble carries the electric bit  $pB = (u > 0)$ ; same-sign charges repel (`moveOneStepAway`), opposite signs aggregate and exchange leader identity [1]. The impulse transferred by a contact is the most literal definition of the coupling; the force-branch statistics that a force law would need are measured and reported in Section 6. Fitting a full  $F(r)$  law, however, meets a structural obstruction: under the single scenario the two-bubble contact is sticky (Section 7), so a clean pre/post force law would require separated sources that weak coupling does not provide.

### 3.2 Operational charge and the discrete Gauss law

The second step defines charge operationally: not as a stored bit, but as the flux of active cells carrying the electric bit through a closed shell. The plan is to verify a discrete Gauss law

$$\Phi(r) = \sum_{\text{active } |x|=r} pB(x) \propto Q_{\text{int}}, \quad (1)$$

which gives the automaton its analogue of  $\varepsilon_0$  and turns the electric force into a flux-measured coupling. The flux census itself is deferred: it requires closed shells around separated charges, which the sticky-contact regime does not supply at these parameters.

## 4 From sieve throughput to the fine-structure constant

### 4.1 The coupling ratio

The third step defines the electromagnetic coupling  $\alpha$  as a ratio of measured events, not as a ratio of stored constants:

$$\alpha(L, S) = \frac{\langle \text{impulse transferred} \rangle}{\langle \text{electric contacts} \rangle}, \quad (2)$$

where the sieve throughput  $P(u, S) \approx u/S$  controls how many overlapping active cells enter the interaction branch [1]. The goal is to show that  $\alpha(L, S)$  is independent of the sieve modulus  $S$

inside the window where the channel opens, and to study its dependence on the lattice size  $L$ . The measured verdict is reported in Sections 6–7: the ratio is resonance-dominated in  $S$  and non-stationary in run length.

## 4.2 Eliminating the dials

The reference implementation sets the wave amplitude seed  $U_0 = 2048$  and the sieve modulus  $S = 16384$  by hand [1]. The plan is to express these constants in terms of the structural integers of the model ( $L$ ,  $W = 3L^2$ ,  $R_{\max} = L/2$ ,  $9L$ ,  $\phi_{\text{full}} = 2R^2$ ) and to test whether the measured ratio (2) converges as  $L$  grows, so that the prediction is a limit rather than a dial setting. Section 7 reports the single- $S$  scan and the  $S$ -averaging attempt that close this route.

## 4.3 Confronting 1/137

The decisive test is numerical: does  $\lim_{L \rightarrow \infty} \alpha(L, S)$  converge to a rational number, and does that number agree with the empirical 1/137.035999084? A clean convergence to 1/137 is the target result; a convergence to any other simple rational is a falsification of the claim that the fine-structure constant is determined by this automaton. Either outcome is publishable. The extrapolation is anchored by the maximum- $L$  and anisotropic-sublattice runs described in Section 5.1; the pre-registered candidate definitions are listed in Section 4.4.

## 4.4 Candidate definitions of $\alpha$

Following the measurement-first methodology of [1], the coupling is measured, not postulated. We pre-register the following definitions, each computable with the existing instrumentation, before any of them is compared with the empirical 1/137.035999177: a “dance on the computer” in which the number comes out, not in which it is put in secretly [7].

1. **Sieve throughput average.**  $\alpha_A = \langle P(u, S) \rangle$  over all active cells of a light frame, with the exact gate probability  $P(u, S) = g[u/g]/S$ ,  $g = \gcd(u, S)$  [1].
2. **Conditional contact ratio.**  $\alpha_B$  as in (2). The reference run already yields the proto-value  $18/6996 \approx 1/388.7$  [1]; the task is its  $L$ -dependence.
3. **Lattice Wyler ratio.**  $\alpha_C$ : a ratio of integer “volumes” forced by the rule—shell counts,  $W = 3L^2$ ,  $\phi_{\text{full}} = 2R^2$ , the  $9L$  election seed—in the spirit of a volume ratio, but with no adjustable radius [2, 3].
4. **Magnetic/electric occupancy.**  $\alpha_D = \langle sB \rangle / \langle pB \rangle$  over a light frame: transverse (magnetic) versus in-phase (electric) bit populations, both measured.
5. **Bound-pair fine structure.**  $\alpha_E$ : the transverse-to-radial energy ratio of a bound pair, computed from the reconstruction pair  $u^2 + v^2 = R^4$ ; only second-order ratios such as  $\langle v^2 \rangle / \langle u^2 \rangle$  are expected to be small.
6. **Self-consistent sieve.**  $\alpha_F = U_0/S^*$ , with  $S^*$  the fixed point of the normalized throughput map  $S \mapsto \langle g[u/g] \rangle(S)$ , or the modulus at which the pair-formation probability crosses 1/2.
7. **Combinatorial state-space ratio.**  $\alpha_G$ : the fraction of the  $2^6$  charge words, or of the  $\phi_{\text{full}}$  phase slots, that participate in pair formation—an Eddington-style count, kept in the list only to be ruled out [4].

The acceptance criterion is deliberately strict: a definition is accepted only if the measured ratio converges to  $1/137.035999177$  to at least eight digits as  $L$  grows, with the definition fixed in advance. A match to only a few digits (as in the historical derivations [5, 6]) is numerology, not a derivation.

## 5 Experiments and configuration

A single headless CPU harness (`alpha_probe`, shipped with the source) implements all probes reported here. Controlled two-bubble runs place two sources  $SEP$  cells apart with complementary charges and opposite unit momenta on the  $x$  axis; the canonical mode allocates the full  $W = 3L^2$  Platonic seed with all sources superposed. Runs use the sieve modulus  $S = 16384$  unless the modulus is the control variable (the  $S$ -scan of Section 7 uses  $S = 128 \dots 16384$ ), 200 light frames (100 at  $L = 13$  for wall-time budget), and the reference geometry  $SEP = 4$ . Each run logs the convolution counters, the per-frame active-cell census, and the realized throughput; the raw CSVs of the  $S$ -scan and the harness itself are shipped with the source.

### 5.1 Lattice budget: maximum $L$ and anisotropic sublattices

Memory and wall time scale with the cell count  $BLOCK \approx L^3W = 3L^5$  (about 64 bytes per cell, so the reference implementation reaches roughly 700 MB at  $L = 31$ ,  $W = 364$  [1]). Two budget rules follow.

**Maximum  $L$ .** Each scenario should be run at the largest lattice side that fits in memory, over the grid  $L = 7, 9, 11, \dots$  up to that limit. The coupling ratio (2) and the force law of Section 3.1 must be studied as functions of  $L$ ; the  $L \rightarrow \infty$  extrapolation of Section 4.3 is the only way to separate genuine predictions from finite-lattice artifacts.

**Anisotropic sublattice.** When a cubic box at the desired  $L$  does not fit in memory, a sublattice  $L \times \frac{L}{16} \times \frac{L}{16}$  with periodic boundary conditions stretches the same cell budget along the scattering axis, cutting the volume by a factor  $16^2 = 256$  and allowing a much larger  $L$ . The two-bubble experiments live on the long axis; the two short axes only need to be wider than the interaction window. This requires generalizing the allocator from the cubic  $EL^3$  to three edge lengths, and the spherical-cavity and shell-counting diagnostics of [1] must be re-validated on the short axes (the cubic torus already exhibits integer-shell anisotropy at  $L \geq 11$  [1]).

## 6 Results

### 6.1 Validation: reproduction of the reference null result

A headless CPU harness (`alpha_probe`, shipped with the source) replays the controlled two-bubble scattering of [1] at the reference parameters ( $L = 7$ ,  $SEP = 4$ , 200 light frames, sieve modulus 16384, complementary charges). The convolution counters reproduce the manuscript exactly:

$$\text{active-passes} = 6996, \quad \text{s2B-passes} = 18, \quad \text{pairs-formed} = 0,$$

i.e. a realized overlap throughput of  $18/6996 \approx 1/388.7$ , the quantitative null result of [1]. Bit-for-bit reproduction is the first check any measurement must pass: the second paper is only as good as its harness.

### 6.2 Candidate $\alpha_A$ : first lattice sweep

The first pre-registered candidate, the frame-averaged gate throughput over the active population, is measured at fixed geometry ( $SEP = 4$ ) and fixed modulus ( $S = 16384$ ):

Table 1: Candidate  $\alpha_A = \langle P(u, S) \rangle$  over the active population, frame-averaged (convolution counts are cumulative over the run).

| $L$ | frames | active passes | $s2B$ passes | $\alpha_A$  | $1/\alpha_A$ |
|-----|--------|---------------|--------------|-------------|--------------|
| 7   | 200    | 6996          | 18           | 0.004220105 | 236.96       |
| 9   | 200    | 9300          | 0            | 0.004265223 | 234.45       |
| 11  | 200    | 11920         | 40           | 0.009991230 | 100.09       |
| 13  | 100    | 7392          | 0            | 0.010428519 | 95.89        |

Two facts follow. First, the harness is consistent with the closed-form gate probability of [1]. Second, and decisive for the program:  $1/\alpha_A$  is *not* constant in  $L$  (236.96, 234.45, 100.09, 95.89). Under the pre-registered acceptance criterion of Section 4.4 this candidate, in this crude form, is *not* a derivation of the fine-structure constant. The non-monotonicity mirrors the non-monotonicity in the sieve modulus reported in [1] (discrete resonances of the overlap), and the fixed  $SEP = 4$  while  $L$  grows changes the collision geometry run to run. A fifth data point from the canonical superposed seed ( $L = 5$ ,  $W = 75$ , 400 frames) gives  $1/\alpha_A = 272.9$ , again frame-dependent and far from  $1/137.036$ . Both the realized ( $1/388.7$ ) and the theoretical ( $1/237$  at  $L = 7$ ) throughputs are nevertheless of the same order as  $1/137.036$ , so the search space is not empty; the remaining candidates of Section 4.4 and the  $SEP \propto L$  scaling are the next measurements.

### 6.3 Candidate $\alpha_D$ : rejected in the two-bubble geometry

The same runs sample the magnetic/electric bit populations  $\langle sB \rangle$  and  $\langle pB \rangle$  at each light frame. The candidate is unstable and non-monotonic ( $\langle sB \rangle / \langle pB \rangle$  by total counts: 0, 3.44, 0.56, 0.81 at  $L = 7, 9, 11, 13$ ). At  $L = 7$  the transverse bit is never set in the controlled two-bubble configuration, i.e. the magnetic channel is not populated there; for  $L \geq 9$  the per-frame ratios are dominated by frames with tiny denominators. Candidate  $\alpha_D$  as defined in Section 4.4 is therefore not measurable in the two-bubble geometry. The canonical Platonic-seed configuration ( $L = 5$ ,  $W = 3L^2 = 75$ , 400 light frames) likewise never populates the magnetic channel ( $\langle sB \rangle = 0$ ), and with all source centres superposed it produces no convolution overlap at all ( $conv\_calls = 0$ ): the sources do not separate within 400 frames in the headless default. The only runs so far in which  $sB > 0$  are the two-bubble geometries at  $L = 9, 11$ , where the ratio is unstable. Observability of  $\alpha_D$  therefore requires separated sources, which the single-scenario dynamics produces from suitably separated initial conditions (as in the two-bubble probe) or only after far longer canonical evolution.

### 6.4 The remaining four candidates: compact report

Candidates  $\alpha_E$ ,  $\alpha_C$ ,  $\alpha_F$  and  $\alpha_B$  complete the pre-registered battery. Their definitions are fixed in Section 4.4; the outcomes are:

Table 2: Report card of the six pre-registered candidates under the strict criterion of Section 4.4 (convergence to  $1/137.035999177$  to at least eight digits).

| Candidate                                                        | Outcome                                                              |
|------------------------------------------------------------------|----------------------------------------------------------------------|
| $\alpha_A$ : $\langle P(u, S) \rangle$ frame-averaged            | $1/237 \dots 1/96$ ( $L = 7 \dots 13$ ); $1/273$ canonical $L = 5$   |
| $\alpha_B$ : force events per contact                            | collapse/gate = 1; reduces to $1/388.7$                              |
| $\alpha_C$ : Wyler-on-lattice, integer ratios                    | $\rightarrow 1/137.03608$ (six digits); best ratio 135.0 at $L = 15$ |
| $\alpha_D$ : $\langle sB \rangle / \langle pB \rangle$           | 0 (weak coupling); $\sim 1$ (open channel)                           |
| $\alpha_E$ : $\langle pol_v^2 \rangle / \langle pol_u^2 \rangle$ | $\rightarrow 2$ exactly (analytic, $R \rightarrow \infty$ )          |
| $\alpha_F$ : $U_0/S^*$ , sieve fixed point                       | $S^* = \langle u \rangle$ (48, 38) $\rightarrow 1/23, 1/54$          |

- $\alpha_E$ . The transverse-to-radial energy ratio of the reconstructed pair is a geometric factor: the ratio of sums over the uniform phase  $j \in [0, 2R)$  converges to 2 as  $R \rightarrow \infty$  (verified to  $R = 1000$ ; the continuous circle limit is exact). The mean-of-ratios variant is ill-defined ( $u = 0$  on pure-transverse cells at even  $R$ ). A related degeneracy explains the  $\alpha_D$  nulls:  $R = R_{\max} - 2 = L/2 - 2$ , so  $L = 7$  gives  $R = 1$  and the reconstruction collapses to  $pol_v \equiv 0$ .
- $\alpha_C$ . Feeding the lattice value of  $\pi$  (integer sphere-point counts) into Wyler’s volume expression converges to Wyler’s numerology  $1/137.036082$ —six digits, below the criterion; no curated structural-integer ratio approaches 137.036 at  $L \leq 31$  (closest 135.0 at  $L = 15$ ).
- $\alpha_F$ . The fixed point  $S^*$  of the normalized throughput map equals the run’s mean active amplitude  $\langle u \rangle$  (48 at  $L = 7$ , 38 at  $L = 9$ ), so  $\alpha_F = U_0/S^* \sim 1/50$  is geometry-locked.
- $\alpha_B$ . Read-only counters at the force-branch decisions show that every gate-passing contact collapses (reissues): collapse/s2B = 1.0 at the reference and 0.995 with the channel open; repulsion is never reached in these probes. The only force of the model is sieve-gated collapse, so  $\alpha_B$  reduces to the realized throughput  $1/388.7$ .

All six pre-registered definitions fail the criterion. This is the primary quantitative result of the paper: at the accessible scales, none of the natural ratio observables of the automaton reproduces the empirical fine-structure constant.

## 7 Discussion

The campaign produced four structural findings and one methodological conclusion.

**Structural findings.** (i)  $L = 7$  is degenerate for the emergent polarization pair:  $R = R_{\max} - 2 = 1$ , so the reconstructed circle collapses and  $pol_v \equiv 0$ ; every earlier  $\alpha_D = 0$  null at  $L = 7$  is a consequence, not a signal. (ii) The two-bubble contact is “sticky”: the adiabatic (non-collapse) branch drifts sources toward each other at every shell coincidence, so isolated bubbles never separate at weak coupling and no clean pre/post collision window exists. (iii) The initial momentum magnitude is inert: sources advance one cell per tick regardless of  $|m|$ , so the emergent maximum information speed cannot be exceeded by construction. (iv) The only force of the model is sieve-gated collapse: every event that passes the  $s2B$  gate reissues; repulsion is structurally unreachable in the probes tested.

**Methodological conclusion.** Six pre-registered candidate definitions, six clean falsifications, none at the  $1/137.036$  level. The negative battery is exactly what the protocol of Section 4.4 was designed to produce: definitions fixed in advance, measured, and rejected under a strict criterion, in the spirit of the historical caution of [5, 6, 7]. It says that within this model, at  $L \leq 31$ , the fine-structure constant does not reside in any ratio of raw emergent observables (gate throughput, magnetic occupancy, reconstruction energy ratio, integer volumes, sieve fixed point). Whether  $\alpha$  requires larger lattices, different initial conditions, or a genuinely different functional of the rule remains open; the search space is bounded but not empty—the realized and theoretical throughputs are both within a factor  $\lesssim 3$  of  $1/137.036$ .

**Bypassing the dial.** Because every candidate inherits the hand-set sieve modulus  $S = 16384 = 2^{14}$ , we attempted to remove the dial in two stages. (i) A single- $S$  scan of the realized throughput at  $L = 7$  is a resonance comb (open channel  $\sim 1$  at  $S \lesssim 10^2$ , crossing  $1/137$  at  $S = 4800$ , collapsing to  $1/437$  at  $S = 5600$ ); the crossing is not  $L$ -stable ( $L = 9$ ,  $S = 4800$  gives  $1/56$ ), so no choice of  $S$  is structural. (ii) Averaging the realized throughput over the modulus grid ( $S = 128 \dots 16384$ ) yields statistics that agree between  $L = 7$  and  $L = 9$  (median  $\sim 1/190$ ) but drift to  $\sim 1/100$  at  $L = 11, 13$ . A frame-count control at  $L = 13$  then shows why: the sieve passes are bursty—they fire when the shell phase coincides—so the realized ratio

changes by factors up to five when the run length doubles ( $S = 5248$ :  $1/241$  at 100 frames vs.  $1/44$  at 200). The realized-overlap throughput is therefore not stationary in time at accessible run lengths, and no  $S$ -averaged count ratio is a stable observable either. The dial cannot be removed by choice or by averaging; only the analytic expected-value observables (the closed-form  $P(u, S)$  averaged over the measured amplitude spectrum) or far longer runs remain. The negative battery of Section 6 therefore stands with an additional methodological control: every count-ratio coupling of this automaton is a discrete resonance in the sieve modulus and in time, not a smooth constant.

## 8 Conclusion

We conducted a systematic, pre-registered search for the fine-structure constant in a concrete “it from bit” automaton: a single-scenario lattice dynamics in which charge, polarization, and a sieve-gated force all emerge from a bit-local rule. Six candidate definitions—fixed in advance, measured with a bit-identical harness, and judged under a strict convergence criterion—all fail to reproduce  $1/137.035999177$  at the accessible scales. Four structural properties of the model were established along the way: the  $L = 7$  degeneracy of the polarization pair, the sticky two-bubble contact, the inexceedable maximum information speed, and the collapse-dominated character of the only force the model possesses.

The negative result is the contribution. It demonstrates the protocol working end to end: in a domain—computational derivations of dimensionless physical constants—where numerology has historically been the norm [5, 6, 7], a concrete automaton now reports six clean falsifications under definitions fixed before measurement. Whether the fine-structure constant lives in this model at all, and if so under which functional of its rule, remains open; what is no longer open is that it is not any of the six natural ratios this paper measured.

## References

- [1] A. Furtado Neto, “It from bit: a concrete attempt,” preprint. Available at [https://www.researchgate.net/publication/372560046\\_It\\_from\\_bit\\_-\\_a\\_concrete\\_attempt](https://www.researchgate.net/publication/372560046_It_from_bit_-_a_concrete_attempt).
- [2] A. Wyler, “Sur la constante de structure fine,” *Comptes Rendus de l’Académie des Sciences*, Sér. A, vol. 269, pp. 743–745, 1969; and “Les groupes des potentiels de Coulomb et de Yukawa,” *ibid.*, vol. 271, pp. 186–188, 1971.
- [3] B. Robertson, “Wyler’s expression for the fine-structure constant  $\alpha$ ,” *Physical Review Letters*, vol. 27, p. 1545, 1971.
- [4] A. S. Eddington, *Fundamental Theory*. Cambridge, UK: Cambridge University Press, 1946.
- [5] H. Kragh, “Magic number: A partial history of the fine-structure constant,” *Archive for History of Exact Sciences*, vol. 57, no. 5, pp. 395–431, 2003.
- [6] I. J. Good, “A quantal hypothesis for hadrons and the judging of physical numerology,” in *Disorder in Physical Systems*, G. R. Grimmett and D. J. A. Welsh, Eds. Oxford, UK: Oxford University Press, 1990, p. 141.
- [7] R. P. Feynman, *QED: The Strange Theory of Light and Matter*. Princeton, NJ: Princeton University Press, 1985.